230X Final Presentation

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October 5th 2026

Section 2

Final Simulation: Group Algorithm

Motivation: What We Saw in Practice

Observation from the practice simulations

Prices were repeatedly pushed hard in one direction: what looked like **market manipulation** drove large, persistent price run-ups. Initial MM Algorithm was performing poorly because of this.

Problem for a market maker

- Quotes both sides, earns the spread
- During a push, gets filled on the losing side
- Builds inventory *against* the move

Our answer: a momentum taker

- Market orders only, no resting quotes
- Detects the push and **follows it**
- Exits when the move fades or reverses

Thesis

If someone is pushing the price, ride the move instead of providing liquidity into it.

Signal: EMA Crossover

Each second, on the mid price m_t:

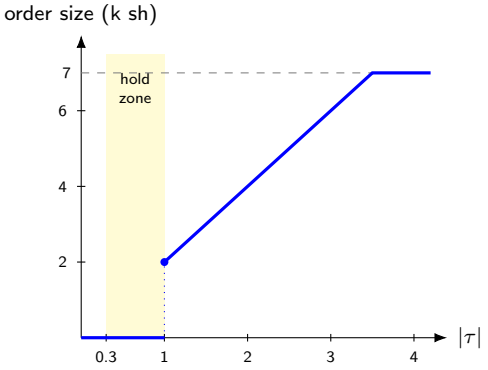
$$\mathrm{EMA}_t^f = \mathrm{EMA}_{t-1}^f + \alpha_f\,(m_t - \mathrm{EMA}_{t-1}^f), \qquad \alpha_f = \tfrac{2}{3+1} = 0.50$$
$$\mathrm{EMA}_t^s = \mathrm{EMA}_{t-1}^s + \alpha_s\,(m_t - \mathrm{EMA}_{t-1}^s), \qquad \alpha_s = \tfrac{2}{12+1} \approx 0.15$$

Trend strength (in \$0.01 ticks)

$$\tau_t = \frac{\mathrm{EMA}_t^f - \mathrm{EMA}_t^s}{\text{tick size}}$$

Sign of τ gives direction (long / short); $|\tau|$ gives conviction and size.

Total Position Sizing



- **Entry**: $|\tau| \geq 1.0$
  size $= \min(7000,\ 2000\,|\tau|)$, in the direction of τ
- **Exit**: $|\tau| \leq 0.3$ or trend flips sign $\Rightarrow$ flatten
- **Hold**: $0.3 < |\tau| < 1.0$, same sign $\Rightarrow$ keep position

Two gates prevent flipping in and out around one threshold.

Budget: $3 \times 7{,}000 = 21{,}000 < 25{,}000$

Execution and Risk Controls

Execution

- Market orders, max 5,000 shares each
- Skip changes under 500 shares
- 0.3 s gap per ticker (RIT position lag)
- Retries on HTTP 429

Risk

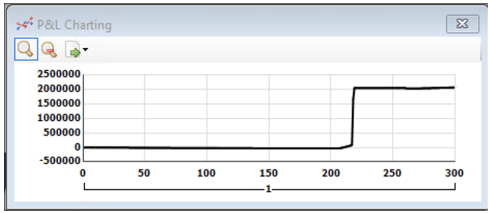
- **Stop**: loss $>$ 4 ticks/share $\Rightarrow$ exit, 5 s cooldown
- **Drawdown breaker**: \$1,500 off peak $\Rightarrow$ flatten, pause 30 s
- **End of case**: flatten in last 5 ticks

Results

Final simulation

Phase	What happened	P&L
1. No clear trend	Algo losing on noise	−$30k
2. Algo off	Manual stop to cap the loss	−$30k
3. Price jumps	Algo back on, rides the push for ∼30 ticks	**+$2.06M**

Extra 5-minute simulation: no more trend, the algo lost **$8k**



Key observation

The edge comes entirely from the manipulation-driven trend: huge payoff when it happens, steady bleed from spread and fees when it does not.

Section 5

Final Simulation: Market-Making Algorithm

Test Sessions: Our Market Maker Made Money Only on RY

Test-session P&L by stock ($k)

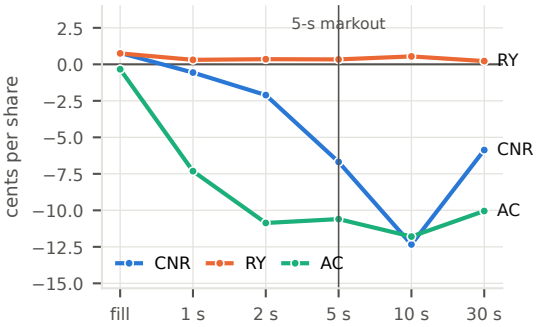
Session	CNR	**RY**	AC
Sep 30 (old engine)	−3.4	**+6.0**	−0.6
Oct 1 afternoon (old engine)	−7.7	−0.2	−2.0
Oct 1 evening, run 1	−14.3	**+37.4**	−13.6
Oct 1 evening, run 2	−13.0	**+20.5**	−653.1*
Oct 2 test	−42.2	**+10.1**	−65.7

Fines excluded (a since-fixed fill-tracking bug). * AC squeeze: $30 → $46 → $29 → $48.

Two risks for a passive market maker

- **Inventory risk**: between the two fills we hold a position. Oct 2: AC rose 18% while we held up to 11,400 shares.
- **Adverse selection**: we get filled just before the price moves against us. 5 s after a fill: CNR −6.7c, AC −10.6c per share; RY +0.3c.

P&L of a passive fill, h seconds later



Oct 2 test, same engine on all three stocks. Spread captured + fee − price move, share-weighted over passive fills.

Decision for the final

Quote RY only. CNR and AC are watched, never quoted.

How the Algorithm Works

1. Measure live

Every second, per stock:
spread, volatility, reversion of 5-s moves (ρ_5),
5-s markouts of our own fills

2. Price

Break-even distance: d_{BE} = measured adverse move − maker fee

Quote at the most aggressive level clearing it (improve, join or behind)

3. Size & inventory

Quotes lean against the position

Adding side shrinks to 0 at the cap

Cut if over the cap or a trend is measured

4. Limits & close

Every order checked as if all resting orders fill (500-share buffer)

Market flatten at tick 296

no fitted parameters | **vs adverse selection** | **vs inventory risk** | **vs limit fines ($1/share)**

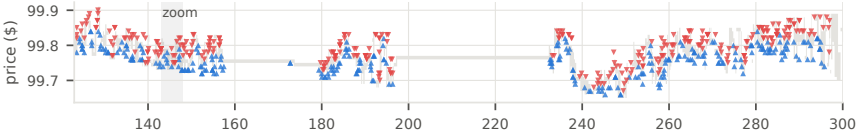
$$f = m + \rho_5\,(m - m_{-5\,\mathrm{s}}), \qquad r = f - \gamma\,\frac{\text{position}}{\text{cap}}, \qquad \text{bid} \le r - d_{BE}, \quad \text{ask} \ge r + d_{BE}$$

Fair value f leans toward reversion when $\rho_5 < 0$ (estimated live, shrunk to 0 without evidence); reservation price r skews quotes to work off inventory.

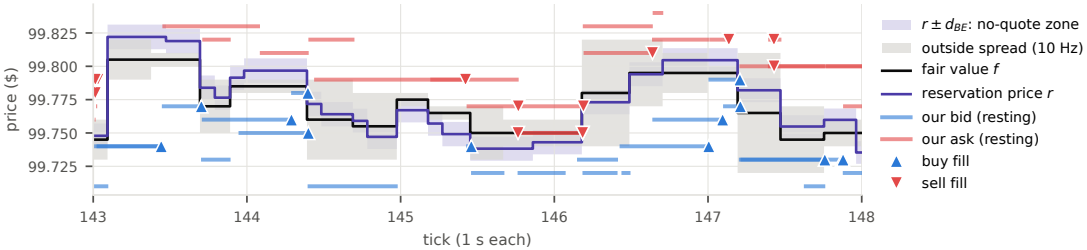
Final Simulation: +$16,323 on RY

$16,323	1.26M	92%	2.95c	$0
net P&L = RIT	RY shares traded	passive fills	sold above bought	fines, flat at the end

RY, whole final: every fill against the outside quotes



Zoom, ticks 143-148: each line is one of our orders, from post to cancel or fill



Net P&L = spread captured +$18,559, inventory −$60, fees −$2,176, fines $0. f, r, d_{BE} logged once a second.